

Query Results

In this section, we will learn how to control the results output of any query.

Sorting Results

We can sort the results of any query using **ORDER BY** and specifying the column to sort by:

Query Results			
mysql> SELECT * FROM logins ORDER BY password;			
+-----+-----+-----+-----+ id username password date_of_joining +-----+-----+-----+-----+			
2 administrator admin_p@ss 2020-07-02 11:30:50			
3 john john123! 2020-07-02 11:47:16			
1 admin p@ssw0rd 2020-07-02 00:00:00			
4 tom tom123! 2020-07-02 11:47:16			
+-----+-----+-----+-----+			
4 rows in set (0.00 sec)			

By default, the sort is done in ascending order, but we can also sort the results by **ASC** or **DESC**:

Query Results			
mysql> SELECT * FROM logins ORDER BY password DESC;			
+-----+-----+-----+-----+ id username password date_of_joining			
+-----+-----+-----+-----+			
4 tom tom123! 2020-07-02 11:47:16			
1 admin p@ssw0rd 2020-07-02 00:00:00			
3 john john123! 2020-07-02 11:47:16			
2 administrator admin_p@ss 2020-07-02 11:30:50			
+-----+-----+-----+-----+			
4 rows in set (0.00 sec)			

It is also possible to sort by multiple columns, to have a secondary sort for duplicate values in one column:

Query Results			
mysql> SELECT * FROM logins ORDER BY password DESC, id ASC;			
+-----+-----+-----+-----+ id username password date_of_joining			
+-----+-----+-----+-----+			
1 admin p@ssw0rd 2020-07-02 00:00:00			
2 administrator change_password 2020-07-02 11:30:50			
3 john change_password 2020-07-02 11:47:16			
4 tom change_password 2020-07-02 11:50:20			
+-----+-----+-----+-----+			
4 rows in set (0.00 sec)			

LIMIT results

In case our query returns a large number of records, we can **LIMIT** the results to what we want only, using **LIMIT** and the number of records we want:

Query Results				
mysql> SELECT * FROM logins LIMIT 2;				
+---+-----+-----+-----+-----+ id username password date_of_joining +---+-----+-----+-----+-----+				
1 admin p@ssw0rd 2020-07-02 00:00:00				
2 administrator adm1n_p@ss 2020-07-02 11:30:50				
+---+-----+-----+-----+-----+				
2 rows in set (0.00 sec)				

If we wanted to **LIMIT** results with an offset, we could specify the offset before the **LIMIT** count:

Query Results				
mysql> SELECT * FROM logins LIMIT 1, 2;				
+---+-----+-----+-----+-----+ id username password date_of_joining +---+-----+-----+-----+-----+				
2 administrator adm1n_p@ss 2020-07-02 11:30:50				
3 john john123! 2020-07-02 11:47:16				
+---+-----+-----+-----+-----+				
2 rows in set (0.00 sec)				

Note: the offset marks the order of the first record to be included, starting from 0. For the above, it starts and includes the 2nd record, and returns two values.

WHERE Clause

To filter or search for specific data, we can use conditions with the **SELECT** statement using the **WHERE** clause, to fine-tune the results:

Code: **sql**

```
SELECT * FROM table_name WHERE <condition>;
```

The query above will return all records which satisfy the given condition. Let us look at an example:

Query Results				
mysql> SELECT * FROM logins WHERE id > 1;				
+---+-----+-----+-----+-----+ id username password date_of_joining +---+-----+-----+-----+-----+				
2 administrator adm1n_p@ss 2020-07-02 11:30:50				
3 john john123! 2020-07-02 11:47:16				
4 tom tom123! 2020-07-02 11:47:16				
+---+-----+-----+-----+-----+				
3 rows in set (0.00 sec)				

The example above selects all records where the value of **id** is greater than 1. As we can see, the first row with its **id** as 1 was skipped

from the output. We can do something similar for usernames:

Query Results				
mysql> SELECT * FROM logins where username = 'admin';				
+-----+-----+-----+-----+-----+ id username password date_of_joining +-----+-----+-----+-----+-----+ 1 admin p@ssw0rd 2020-07-02 00:00:00 +-----+-----+-----+-----+-----+ 1 row in set (0.00 sec)				

The query above selects the record where the username is **admin**. We can use the **UPDATE** statement to update certain records that meet a specific condition.

Note: String and date data types should be surrounded by single quote (') or double quotes ("), while numbers can be used directly.

LIKE Clause

Another useful SQL clause is **LIKE**, enabling selecting records by matching a certain pattern. The query below retrieves all records with usernames starting with **admin**:

Query Results				
mysql> SELECT * FROM logins WHERE username LIKE 'admin%';				
+-----+-----+-----+-----+-----+ id username password date_of_joining +-----+-----+-----+-----+-----+ 1 admin p@ssw0rd 2020-07-02 00:00:00 4 administrator adm1n_p@ss 2020-07-02 15:19:02 +-----+-----+-----+-----+-----+ 2 rows in set (0.00 sec)				

The **%** symbol acts as a wildcard and matches all characters after **admin**. It is used to match zero or more characters. Similarly, the **_** symbol is used to match exactly one character. The below query matches all usernames with exactly three characters in them, which in this case was **tom**:

Query Results				
mysql> SELECT * FROM logins WHERE username like '___';				
+-----+-----+-----+-----+-----+ id username password date_of_joining +-----+-----+-----+-----+-----+ 3 tom tom123! 2020-07-02 15:18:56 +-----+-----+-----+-----+-----+ 1 row in set (0.01 sec)				



Connect to Pwnbox

Your own web-based Parrot Linux instance to play our labs.

Pwnbox Location

AU

80ms



Terminate Pwnbox to switch location

Start Instance

∞ / 1 spawns left

Waiting to start...



Questions

Answer the question(s) below to complete this Section and earn cubes!



Cheat Sheet

Target(s): 94.237.59.199:54178

Life Left: 80 minute(s)

Authenticate to 94.237.59.199:54178 with user "root" and password "password"

+ 1 What is the last name of the employee whose first name starts with "Bar" AND who was hired on 1990-01-01?

Submit your answer here...

+10 Streak pts



Submit



Hint



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Cheat Sheet

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Start Instance

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